Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

Tim Dennis
Data Science Center, UCLA Library

2026-01-01

The invisible layer

Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **67%** say their research would be impossible without it.

The people keeping it running are increasingly called **Research Software Engineers (RSEs)** - a growing professional community with no standard institutional home.

Hettrick et al. (SSI, 2014/2022)

We’ve been here before

People in this room helped build the open data movement:

- FAIR principles: from aspiration to institutional expectation
- Data management plans embedded in grant workflows
- Curation, provenance, and stewardship as professional services
- Open science infrastructure: repositories, PIDs, metadata standards
- **FAIR4RS**: FAIR principles now extended to research software

Software is the next frontier — and UC has been showing the way for 50 years: BSD Unix, Ceph, Apache Spark, RISC-V, Ray. The pipeline from university research to global infrastructure is not rare or accidental.

> “The research done in academia was the seed. Foundations provided the soil. But the community was the sun.”

Barker et al. (2022) doi:10.1038/s41597-022-01710-x · Ruff, N. (2026) “The Role of Foundations in Advancing Open Collaboration and Innovation,” UC Open Summit. youtu.be/eBriL3CDNeo

The UC research-to-infrastructure pipeline

UC Berkeley

- Jupyter
- Apache Spark
- RISC-V
- Ray
- BSD Unix

UC Santa Cruz

- Ceph
- **UCSC Genome Browser**

UC San Diego

- **Cytoscape**
- **EEGLAB**
- UCSD Pascal

UCLA

- Processing
- Named Data Networking
- **tz database**

UC San Francisco

- **ChimeraX**
- OpenMM

UC Davis

- sourmash
- **ERPLAB**

Summer 2000. UC Santa Cruz.

PhD student Jim Kent assembled the human genome on 50 borrowed PCs, icing his wrists to keep coding.

He finished **three days** before a corporate competitor’s supercomputer.

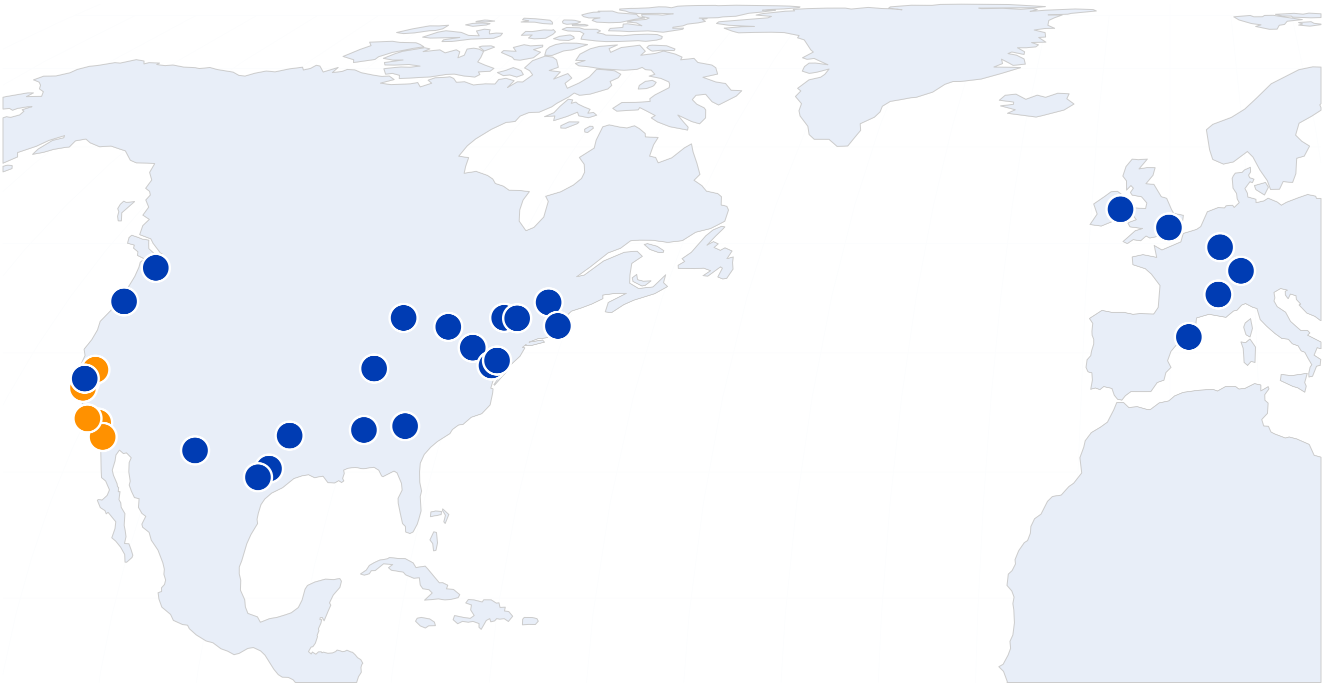
The genome stayed in the public domain.

Kent, J.W. (2002). BLAT — The BLAST-Like Alignment Tool. *Genome Research* 12, 656–664. · genome.ucsc.edu

What is an academic OSPO?

An **Open Source Program Office** supports, governs, and promotes open source within an institution.

- Originated in industry: ~77% of large tech companies have one
- First academic OSPO: Johns Hopkins, 2019
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



🟠 UC OSPO Network · 🔵 CURIOSS member

The UC OSPO Network

A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
- Launched April 2024, funded by the Alfred P. Sloan Foundation
- **$1.85M** in external funding to date
- Lead: UC Santa Cruz, the first OSPO in a large state university system
- Serves 280,000+ students and 25,000+ faculty

Acts as a **neutral convener**: no single campus, company, or grant cycle can pull the work away.

Ruff (2026) UC Open Summit · youtu.be/eBriL3CDNeo

Three thematic areas

Discovery

Mapping who does what across the UC system — repos, contributors, tools, and practices

Sustainability

Keeping projects and communities healthy through licensing, project health, and community support

Education

Coordinated training and learning pathways across campuses

Discovery: knowing what’s there

The 4 Ps framework, applied at system scale:

- **People**: who contributes to OS software at UC?
- **Product**: what has been built?
- **Practice**: how do faculty and staff engage?
- **Perception**: what are contributors’ experiences?

Tools: GitHub pipeline (200,000+ repos scanned; ~52,000 institutionally affiliated across 10 campuses), network-wide survey (294 respondents), UC Open Repository Browser (UC ORB)

Gomez et al. (2025) arxiv:2506.18359 · Scarlett et al. (2025) doi:10.31235/osf.io/p8bx6_v1

Sustainability: keeping it running

58% of UC open source contributors are also maintainers. Not users: stewards.

What they asked for most: sustainability grants, computing infrastructure, learning communities.

#1 challenge: **finding time to write documentation.**

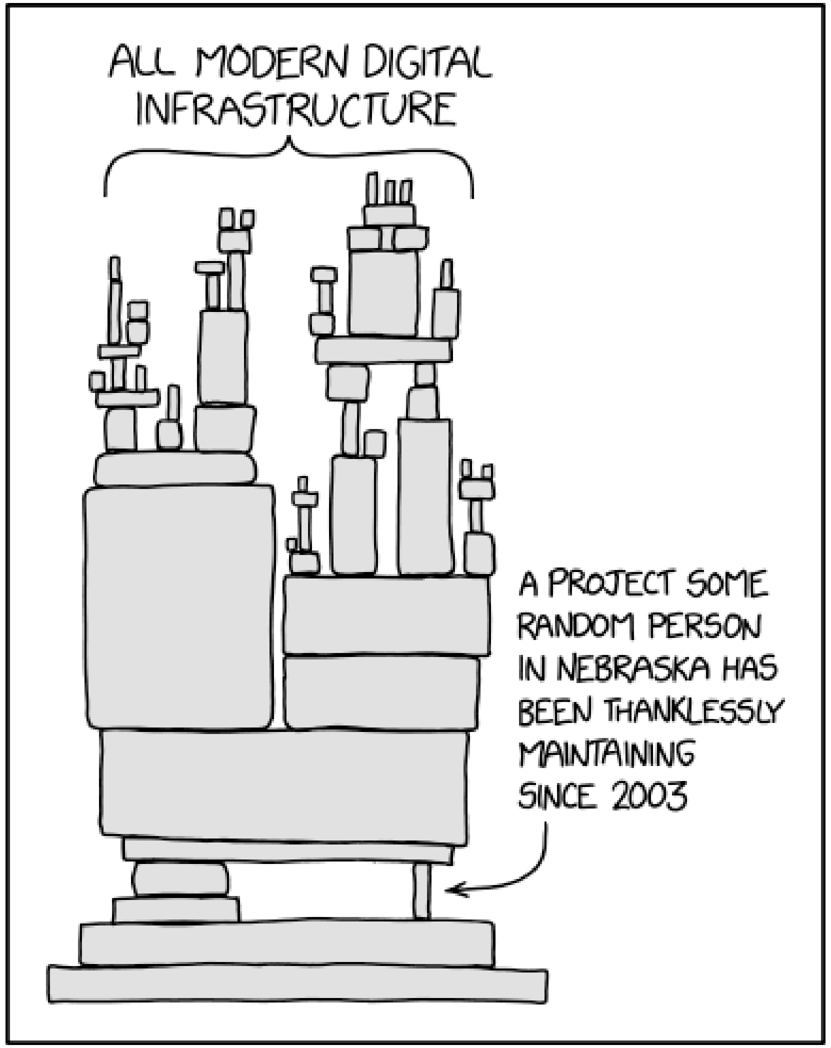
Nationally: **60%** of maintainers are unpaid; **43%** report burnout.

Network services:

- Cross-campus licensing working group (UCOP + tech transfer + IT)
- Project health assessment (OSSPREY)
- Containerization support · Community management

Scarlett et al. (2025) doi:10.31235/osf.io/p8bx6_v1 · Tidelift (2024) State of the Open Source Maintainer Report

The dependency problem



XKCD #2347 ‘Dependency’: a towering stack of blocks labeled ‘All modern digital infrastructure’ balances on a single tiny block labeled ‘A project some random person in Nebraska has been thanklessly maintaining since 2003’. By Randall Munroe.

xkcd #2347 (CC BY-NC 2.5) · UC survey: 58% of contributors are maintainers · Tidelift (2024): 60% unpaid, 43% burnt out

AI and open source: a new pressure

AI code generation is changing what it means to maintain a project.

- **Slop PRs**: AI-generated contributions that look complete but are off-spec, shallow, or carry subtle bugs
- Review burden grows while maintainer capacity stays flat
- The OSS projects that trained the models are now flooded by their output
- Security risk: AI-generated code in dependencies with undetected vulnerabilities

OSPOs are developing responses: contribution disclosure policies, AI-aware code review norms, project-level guidance.

Baltes, Cheong & Treude (2026) arxiv:2603.27249 · GitClear (2024) “Coding on Copilot”

The structural shift

70% of new AI PhDs go straight to industry

(a decade ago: 50/50)

90% of notable AI models come from a handful of companies

Academia is being sidelined from the systems it helped create.

Ruff (2026) UC Open Summit · youtu.be/eBriL3CDNeo

Education: training as infrastructure

Gap analysis → curriculum inventory → learning pathways

- Inventoried existing materials: Carpentries, CodeRefinery, The Turing Way, UC-specific content
- Organized by topic (licensing, sustainability, community) and by role (contributor, maintainer, manager)
- Published at ucospo.net/education
- Coordinated with data literacy programs and Carpentries infrastructure

Governance: intra vs. inter campus

Not everything should be shared. Some things only work at scale.

Intra-campus	Inter-campus
Initial pilots	Platform and tool development
Local relationship-building	Licensing and best practices
Faster, context-specific support	System-wide discovery
Unique institutional context	Field-specific community building

Network governance: OSPO Leadership Group + working groups; GitHub-based project management.

Lessons from two years

What the network model enables that single campuses cannot:

1. **Shared staffing**: community manager, licensing specialist, technical roles
2. **Policy leverage**: the network has standing with UCOP; individual campuses do not
3. **Data at scale**: the GitHub and survey analyses require system-wide scope to mean anything
4. **Equity**: smaller campuses get services they could not build alone
5. **A replicable model**: $1.85M produced something other state systems can follow

Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

What you can do right now

If you advise on data-intensive research, you are probably already helping with code.

The network has already built the resources:

- `README.md` — template + guide (Laura Langdon, UC OSPO)
- `LICENSE` — guide includes UC-approved options
- `CONTRIBUTING.md` — template + guide
- `CODE_OF_CONDUCT.md` — Contributor Covenant + guidance
- `CITATION.cff` + Zenodo release → citable, versioned DOI

Going further:

- Growing a Community Around a Project — UC OSPO
- Sharing Research Software: Reproducible, Citable, and Discoverable — Carpentries Incubator lesson (covers `pixi`, CITATION.cff, Zenodo, discoverability)

Learn more / connect

UC OSPO Network: ucospo.net

- Education site: ucospo.net/education
- Mailing list, Slack, events calendar

Global network: curioss.org — 34 academic OSPOs worldwide and growing

Tim Dennis Data Science Center, UCLA Library dennis.tim@gmail.com